MILITARY STANDARD

AUTOMATED ENGINEERING
DOCUMENT PREPARATION SYSTEM

BOLTS AND STUDS

TO ALL HOLDERS OF DOD-STD-35-31A(MI):

1. THE FOLLOWING PAGES OF DOD-STD-35-31A(MI) HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED.

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
v	30 JUNE 1992	v	23 MARCH 1979
3	30 JUNE 1992	3	23 MARCH 1979
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55	23 MARCH 1979	(Reprinted without change)	
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58	23 MARCH 1979	(Reprinted without change)	

2. RETAIN THIS NOTICE PAGE AND INSERT BEFORE THE TABLE OF CONTENTS.

3. Holders of DOD-STD-35-31A(MI) will verify that page changes and additions indicated above have been entered. The notice page will be retained as a check sheet. This issuance, together with appended pages is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

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AREA EGDS

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TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BOLTS	152005 (CONT)	NAS498	SHEAR
		NAS6203 THRU NAS6220	SHEAR CL TOL
		NAS333 THRU NAS340	100 DEG CL TOL H STR
		NAS501	STABILIZED NMAG CL TOL
		NAS563 THRU NAS572	FULLY THD, FULLY IDENTIFIED HD
		NAS621	ALLOY STL
		NAS673 THRU NAS678	CL TOL HEX HD TITANIUM, .190 TO .500
		NAS1266 THRU NAS1270	LOCK, HEX HD CL TOL
		NAS2005	LOCK, TNSN PROTRUDING HD PULL TYPE
		NAS2105 THRU NAS2112	LOCK, TNSN 100 DEG (AN509) HD
		NAS2205 THRU NAS2212	LOCK, TENSION, PROTRUDING HD, STUMP TYPE
		NAS2306 THRU NAS2312	LOCK, TENSION, 100 DEGREE (AN509) HD, STUMP TYPE
		NAS2406 THRU NAS2412	LOCK, SHEAR PROTRUDING HD PULL TYPE

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MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BOLTS	152005 (CONT)	NAS2506 THRU NAS2512	LOCK, SHEAR, 100 DEGREE HD PULL TYPE
		NAS2605 THRU NAS2612	LOCK, SHEAR PROTRUDING HD, STUMP TYPE
		NAS2705 THRU NAS2712	LOCK, TNSN PULL TYPE
		NAS1578	SHEAR, PAN HD 1200, DEGREE F
		NAS4002	ALLOY STL EXTLY THREADED
		NAS6203 THRU NAS6220	HEX HD, CL TOL ALLOY STL SHORT THD
		NAS6603 THRU NAS6620	HEX HD, CL TOL ALLOY STL L THD
		NAS4004	FSTNR, 6AL-4V TITANIUM ALLOY EXT THD
		NAS6403 THRU NAS6420	HEX HD CL TOL SHORT THD SLFLKG AND NONLKG

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MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BOLTS	152005 (CONT)	NAS6803 THRU NAS6820	HEX HD CL TOL 6AL-4V LG THD SLFLKG AND NONLKG
BOLTS, FIRE BRICK ANCHOR	152006	MIL-B-15382	
BOLTS, HOOK-TURNBUCKLE	152007	FF-T-791	
		MS27593	GENERAL PURPOSE
BOLTS, INTERNAL WRENCHING	152008	MIL-B-7838	160 KSI
		MS20004 THRU	
		MS20024	160 AND 96 KSI
BOLTS, LAG	152009	FF-B-561	
		MS16201	SLTD FLH, CONE PT NMAG
		MS16992	HEX HD, GIMLET PT
		MIL-H-55053	HDW, POLE LINE
		MS35801	POLE LINE, FETTER DRIVE
BOLTS, MACHINE, DOUBLE HEX WASHER HEAD, FULL SHANK	152010	MS9110	CL TOL .190-32
		MS9111	CL TOL .250-28
		MS9112	CL TOL .3125-24
		MS9113	CL TOL .375-24
		MS9933	CL PL .250-28

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MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BOLTS, MACHINE, DOUBLE HEX WASHER HEAD, FULL SHANK	152010 (CONT)	MS9944	CD PL .375-24, DRILLED
		MS9947	CD PL .5625-18, DRILLED
BOLTS, MACHINE, DOUBLE HEX WASHER, PITCH DIAMETER SHANK	152011	MS9208	NI-CAD PL .190-32
		MS9209	NI-CAD PL .250-28
		MS9210	NI-CAD PL .3125-24
		MS9211	NI-CAD PL .375-24
		MS9212	NI-CAD .4375-20
		MS9213	NI-CAD PL .500-20
		MS9214	NI-CAD PL .5625-18
		MS9217	NI-CAD PL .190-32, DRILLED
		MS9218	NI-CAD PL .250-28, DRILLED
		MS9219	NI-CAD PL .3125-24, DRILLED
		MS9220	NI-CAD PL .375-24, DRILLED
		MS9221	NI-CAD PL .4375-20, DRILLED
		MS9222	NI-CAD PL .500-20, DRILLED
		MS9223	NI-CAD PL .5624-18

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BOLTS, MACHINE, DOUBLE HEX WASHER, PITCH DIAMETER SHANK	152011 (CONT)	MS9556	AMS 5731, .190-32
		MS9557	AMS 5731, .250-28
		MS9558	AMS 5731, .3125-24
		MS9559	AMS 5731, .375-24
		MS9560	AMS 5731, .4375-20
		MS9561	AMD 5731, .500-20
		MS9562	AMS 5731, .5625-18
		MS9565	AMS 5731, .190-32, DRILLED
		MS9566	AMS 5731, .250-28, DRILLED
		MS9567	AMS 5731, .3125-24, DRILLED
		MS9568	AMS 5731, .375-24, DRILLED
		MS9569	AMS5731, .4375-20
		MS9570	AMS 5731, .500-20
		MS9574	SIL PLD, .190-32, DRILLED
		MS9575	SIL PLD, .250-28, DRILLED
		MS9576	SIL PLD, .3125-24, DRILLED

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MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BOLTS, MACHINE, DOUBLE HEX WASHER, PITCH DIAMETER SHANK	152011 (CONT)	MS9577	SIL PLD, .375-24, DRILLED
		MS9578	SIL PLD, .4375-20, DRILLED
		MS9579	SIL PLD, .500-20, DRILLED
		MS9580	SIL PLD, .5625-18, DRILLED
		MS9676	AMS 5731, CUPWASHER LKD, .190-32
		MS9677	AMS 5731, CUPWASHER LKD, .250-28
		MS9678	AMS5731, CUPWASHER LKD, .3125-24
		MS9679	AMS 5731, CUPWASHER LKD, .375-24
		MS9680	CAD PLATED, CUPWASHER LKD, .190-32
		MS9681	CAD PLATED, CUPWASHER LKD, .250-28
		MS9682	CAD PLATED, CUPWASHER LKD, .3125-24
		MS9683	CAD PLATED, CUPWASHER LKD, .375-24
		MS9696	AMS 5708, .190-32
		MS9697	AMS 5708, .250-28
		MS9698	AMS 5708, .3125-24

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BOLTS, MACHINE, DOUBLE HEX WASHER, PITCH DIAMETER SHANK	152011 (CONT)	MS9699	AMS 5708, .375-24
		MS9700	AMS 5708, .4375-20
		MS9701	AMS 5708, .500-20
		MS9702	AMS 5708, .5625-18
		MS9705	AMS 5708, .190-32, DRILLED
		MS9706	AMS 5708, .250-28, DRILLED
		MS9707	AMS 5708, .3125-24, DRILLED
		MS9708	AMS 5708, .375-24, DRILLED
		MS9709	AMS 5708, .4375-20, DRILLED
		MS9710	AMS 5708, .500-20,, DRILLED
		MS9711	AMS 5708, .5625-18, DRILLED
		MS9714	SIL PLD, .190-32, DRILLED
		MS9715	SIL PLD, .250-28, DRILLED
		MS9716	SIL PLD, .3125-24, DRILLED
		MS9717	SIL PLD, .375-24, DRILLED

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MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BOLTS, MACHINE, DOUBLE HEX WASHER, PITCH DIAMETER SHANK	152011 (CONT)	MS9718	SIL PLD, .4375-20, DRILLED
		MS9719	SIL PLD, .500-20, DRILLED
		MS9720	SIL PLD, .5625-18, DRILLED
		MS9732	CRES AMS 5643, .190-32
		MS9733	CRES AMS 5643, .250-28
		MS9734	CRES AMS 5643, .3125-24
		MS9741	CRES AMS 5643, .190-32, DRILLED
		MS9742	CRES AMS 5643, .250-28, DRILLED
		MS9743	CRES AMS 5643, .3125-24, DRILLED

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BOLTS, MACHINE, DOUBLE HEX WASHER, PITCH DIAMETER SHANK	152011 (CONT)	MS9750	TITANIUM AMS 4967, .190-32
		MS9751	TITANIUM AMS 4967, .250-28
		MS9759	TITANIUM AMS 4967, .190-32, DRILLED
		MS9760	TITANIUM AMS 4967, .250-28, DRILLED
		MS9761	TITANIUM AMS 4967, .3125-24, DRILLED

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MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BOLTS, MACHINE, DOUBLE HEX WASHER, PITCH DIAMETER SHANK	152011 (CONT)	MS9885	AMS 5616, .190-32
		MS9887	AMS 5616, .375-24
		MS9890	AMS 5616, .500-20
		MS9894	AMS 5616, .190-32, DRILLED
		MS9895	AMS 5616, .250-28, DRILLED
		MS9896	AMS 5616, .3125-24, DRILLED
		MS9897	AMS 5616, .375-24, DRILLED
		MS9898	AMS 5616, .4375-20, DRILLED
		MS9899	AMS 5616, .500-20, DRILLED

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MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BOLTS, MACHINE, DOUBLE HEX WASHER, PITCH DIAMETER SHANK	152011 (CONT)	MS9900	AMS 5616, .5615-18 DRILLED
		MS9914	CAD PLD, .190-32
		MS9917	CAD PLD, .375-24
		MS9918	CAD PLD, .4375-20
		MS9923	CAD PLD, .190-32, DRILLED
		MS21279	CONTROLLED ROOT RAD, .190-32
		MS21280	CONTROLLED ROOT RAD, .250-28
		MS21281	CONTROLLED ROOT RAD, .3125-24

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MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BOLTS, MACHINE, DOUBLE HEX WASHER, PITCH DIAMETER SHANK	152011 (CONT)	MS21282	CONTROLLED ROOT RAD, .375-24
		MS21284	CONTROLLED ROOT RAD, .500-50
		MS21285	CONTROLLED ROOT RAD, .5625-18
		MS21288	CONTROLLED ROOT RAD, .190-32, DRILLED
		MS21290	CONTROLLED ROOT RAD, .3125-24, DRILLED
		MS21291	CONTROLLED ROOT RAD, .375-24, DRILLED
		MS21292	CONTROLLED ROOT RAD, .4375-20, DRILLED
		MS21293	CONTROLLED ROOT RAD, .500-20, DRILLED
		MS21294	CONTROLLED ROOT RAD, .5625-18, DRILLED
BOLTS, MACHINE, HEX HEAD, FULL SHANK	152012	MIL-B-7874	1200 DEGREES F
		MS9489	CRES, .190-32
		MS9490	CRES, .250-28
		MS9491	CRES, .3125-24
		MS9492	CRES, .375-24
		MS9493	CRES, .4375-20
		MS9494	CRES, .500-20
		MS9496	CRES, .625-18
		MS9497	CRES, .750-16

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BOLTS, MACHINE, HEX HEAD, FULL SHANK	152012 (CONT)	MS9500	CRES, .190-32, DRILLED
		MS9501	CRES, .250-28, DRILLED
		MS9502	CRES, .3125-24, DRILLED
		MS9503	CRES, .375-24, DRILLED
		MS9505	CRES, .500-20, DRILLED
		MS9506	CRES, .5625-18, DRILLED
		MS9507	CRES, .625-18, DRILLED
		MS9529	CAD PLD, .190-32, DRILLED
		MS9583	CRES, .190-32, DRILLED 6 HOLES
		MS9584	CRES, .250-28, DRILLED 6 HOLES
		MS9585	CRES, .3125-24, DRILLED 6 HOLES
		MS9586	CRES, .375-24, DRILLED 6 HOLES
		MS9587	CRES, .4375-20, DRILLED 6 HOLES
		MS9588	CRES, .500-20, DRILLED 6 HOLES
		MS9589	CRES, .5625-18, DRILLED 6 HOLES
		MS9590	CRES, .025-18, DRILLED 6 HOLES
		MS9591	CRES, .750-16, DRILLED 6 HOLES
		MS9642	TITANIUM, .190-32, DRILLED

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MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BOLTS, MACHINE, HEX HEAD, FULL SHANK	152012 (CONT)	MS9643	TITANIUM, .250-28, DRILLED
		MS9644	TITANIUM, .3125-24, DRILLED
		MS9645	TITANIUM, .375-24, DRILLED
		MS9646	TITANIUM, .4375-20, DRILLED
		MS9647	TITANIUM, .500-20, DRILLED
		MS9648	TITANIUM, .5625-18, DRILLED
		MS9652	TITANIUM, .250-18
		MS9653	TITANIUM, .3125-24
		MS9654	TITANIUM, .375-24
		MS9655	TITANIUM, .4375-20
		MS9656	TITANIUM, .500-20
		MS9657	TITANIUM, .5625-18
		MS9783	AMS 5643, .190-32
		MS9784	AMS 5643, .250-28
		MS9785	AMS 5643, .3125-24
		MS9786	AMS 5643, .375-24
		MS9787	AMS 5643, .4375-20

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BOLTS, MACHINE, HEX HEAD, PITCH DIAMETER SHANK	152013 (CONT)	MS9630	TITANIUM, .5625-18, DRILLED
		MS9633	TITANIUM, .190-32
		MS9634	TITANIUM, .250-28
		MS9635	TITANIUM, .3125-24
		MS9636	TITANIUM, .375-24
		MS9637	TITANIUM, .4375-20
		MS9638	TITANIUM, .500-20
		MS9639	TITANIUM, .5625-18
		MS9685	NI-CAD PL, .190-32, DRILLED
		MS9686	NI-CAD PL, .250-28, DRILLED
		MS9687	NI-CAD PL, .3125-24, DRILLED

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BOLTS, MACHINE, HEX HEAD, PITCH DIAMETER SHANKANK	152013 (CONT)	MS9805	CRES, .190-32
		MS9806	CRES, .250-28
		MS9807	CRES, .3125-24
		MS9808	CRES, .375-24
		MS9809	CRES, .4375-20
		MS9810	CRES, .500-20
		MS9811	CRES, .5625-18
		MS9812	CRES, .625-18
		MS9813	CRES, .750-16
		MS9816	CRES, .190-32, DRILLED
		MS9817	CRES, .250-28, DRILLED
		MS9818	CRES, .3125-24, DRILLED
		MS9819	CRES, .375-24, DRILLED
		MS9820	CRES, .4375-20, DRILLED
		MS9821	CRES, .500-20, DRILLED
		MS9822	CRES, .5625-18, DRILLED
		MS9823	CRES, .625-18, DRILLED
		MS9824	CRES- .750-16, DRILLED
BOLTS, MACHINE, HEX AND SQUARE HEAD	152014	FF-B-575	BOLT, HEX AND SQUARE
		MS35355	CAP ON ZINC PL
		MS35356	PHOS COATED
BOLTS, MACHINE, SQUARE HEAD	152015	MIL-H-55053	HDW, POLE LINE
		MS35798	GALVANIZED (POLE LINE)

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BOLTS, MACHINE, TENSILE	152016	MIL-B-8831	180 KSI, EXT WP, FLGD HD
		MIL-B-8906	220 KSI, EXT WR, FLGD HD
		MIL-B-8907	260 KSI, EXT WR, FLGD HD
		MS21250	180 KSI, EXT WR, FLGD HD
BOLTS, MACHINE, 12 POINT HEAD	152017	MS9088	CAD PLD, .190-32, DRILLED
		MS9089	CAD PLD, .250-28, DRILLED
		MS9090	CAD PLD, .3125-24, DRILLED
		MS9091	CAD PLD, .375-24, DRILLED
		MS9093	CAD PLD, .500-20, DRILLED
		MS9146	CAD PLD, .190-32
		MS9147	CAD PLD, .250-28
		MS9148	CAD PLD, .3125-24
		MS9149	CAD PLD, .375-24
		MS9150	CAD PLD, .4375-20
		MS9151	CAD PLD, .500-20
		MS9158	BLACK OXIDE, .250-28
		MS9160	BLACK OXIDE, .375-24
		MS9169	BLACK OXIDE, .190-32, DRILLED

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TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BOLTS, MACHINE, 12 POINT HEAD	152017 (CONT)	MS9170	BLACK OXIDE, .250-28, DRILLED
		MS9171	BLACK OXIDE, .3125-24, DRILLED
		MS9172	BLACK OXIDE, .375-24, DRILLED
		MS9173	BLACK OXIDE, .4375-20, DRILLED
		MS9174	BLACK OXIDE, .500-50, DRILLED
		MS9175	BLACK OXIDE, .5625-18, DRILLED
		MS9224	HEAT RES, .5625-18
BOLTS, SELF-LOCKING	152018	MIL-B-45912	BOLT, SLFLKG, HEX HEAD, STEEL
		MS21091	75 AND 125 KSI, FLUSH HEAD
		MS21092	48 AND 80 KSI, FLUSH HD, CRES
		MS21094	75 AND 125 KSI, HEX HD
		MS21095	48 AND 80 KSI, HEX HD, CRES
		MS21096	75 AND 125 KSI, PAN HEAD
		MS21097	48 AND 80 KSI, PAN HD, CRES
		MS21098	160 KSI, 12 POINT HEAD
		MS21099	80 KSI, 12 PT HEAD, CRES
		MS35764	HEX HD, GRADE 8

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TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BOLTS, SELF RETAINING	152019	MIL-B-83050	BOLT, SLFRTNG, IMPEDANCE TYPE
		MS27576	IMPEDANCE TYPE, 95 KSI, HEX HD
		MS27577	IMPEDANCE TYPE, 95 KSI, FL HD
		MS33602	DESIGN AND USAGE REQTS
BOLTS, SHOULDER	152020	MS21301	BOLT, SHOULDER
		MS51946	RIBBED, SPLINED 3/4-INCH
		MS51949	WHEEL MOUNTING, KEYED HEAD
BOLTS, SQUARE NECK	15207	MS16207	CARRIAGE, COP SILICON, NMAG
		MS35754	CTSK (PLOW), CAD OR ZINC PLATED
BOLTS, TEE HEAD	152022	MIL-B-45887	BOLTS, TEE HEAD
		MS9397	CAD PLD, .190-32
		MS9398	CAD PLD, .250-28
		MS9399	CAD PLD, .3125-24
		MS9400	CAD PLD, .375-24
		MS9401	CAD PLD, .4375-20
		MS9402	CAD PLD, .500-20
		MS9432	AMS 5735, .190-32
		MS9433	AMS 5735, .3125-24
		MS9434	AMS 5735, .3125-24
		MS9435	AMS 5735, .375-24

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TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BOLTS, TEE HEAD	152022 (CONT)	MS9436	AMS 5735, .4375-20
		MS9437	AMS 5735, .500-20
		MS90729	CSK HEAD
BOLTS, THIMBLE EYE	152023	MIL-H-55053	HDW POLE LINE
		MS35796	STR, POLE LINE
		MS35797	BENT, POLE LINE
BOLTS, TIBIAL	152024	MIL-B-36772	
BOLTS, TOGGLE	152025	FF-B-588	
BOLTS, TRACK	152026	ANSIB18.10	OVAL NECK
BOLTS, WELDING	152028	MS51933	FINE THREAD
		MS51934	COARSE THREAD
BOLTS, WHEEL MOUNTING	152029	MS51947	SPLINED, .750 INCH
BOLT, TENSION	152030	MS21296	MP35N, 260 KSI FTU, 12 PT EXT WRNG, FLGD HD, 700 DEG F (371.1 DEG C)
STUDS, CONTINUOUS THREAD	152031	MIL-S-1222	HIGH TEMP
		MS16187	B16 ALLOY
STUDS, LOCKED IN	152032	MIL-S-45909	RING LOCKED
		MS51989	RING LKD, SERR
		MS51992	RING LKD, SERR, HIGH STRENGTH
		MIL-S-45915	KEY LKD, SERR
		MS51833	KEY LKD, LTWT
		MS51834	KEY LKD, HD

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TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
STUDS, LOCKED IN	152032 (CONT)	MIL--S-45933	KEYRING LDK, 125 AND 160 KSI
		MS51999	RING LKD, SERR, USAGE LIMITATIONS
STUDS, PLAIN	152033	FF-S-1362 MS51864	STUDS, PLAIN GENERAL PURPOSE
STUDS, SHOULDERED	152034	MS51950	WHEEL MOUNTING
STUDS, SHOULDERED, STEPPED	152035	MS51951	WHEEL MOUNTING
STUDS, STEPPED, 1.5 DIAMETER ENGAGEMENT	152037	AN150501 THRU AN150800	.250-20 X .190-32
		ANA-BUL-337	.3125-18 X .250-28
		ANA-BUL-337	.375-16 X .3125-24
		AN151401 THRU AN151700	.4375-14 X .375-24
		AN152601 THRU AN152900	DRILLED .250-20 X .190-32
		AN152901 THRU AN153200	DRILLED, .3125-18 X .250-20

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TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
STUDS, STEPPED, 1.5 DIAMETER ENGAGEMENT	152037 (CONT)	MS17294	CONTROLLED ROOT RAD .3125-18 X .250-28
		MS17295	CONTROLLED ROOT RAD .375-16 X .3125-24
		MS17296	CONTROLLED ROOT RAD .4375-14 X .375-24
STUDS, STEPPED, 2.0 DIAMETER ENGAGEMENT	152039	AN154701 THRU AN155000	.250-20 X .190-32

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TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
STUDS, STEPPED, 2.0 DIAMETER ENGAGEMENT	152039 (CONT)	MS9837	DRILLED, .4375-14 X .375-24
STUDS, STEPPED, NECKED, 2 DIAMETER ENGAGEMENT	152040	AN162501 THRU AN162800	DRILLED, .4375-14 X .375-24
STUDS, STRAIGHT	152041	MIL-S-24287	BOLT-STUD, NUTS ALLOY STEEL (FOR SERVICE TO 700 DEG F AND FATIGUE APPLS, ISSUE CONTD)

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TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
STUDS, STRAIGHT	152041 (CONT)	MIL-S-24354	BOLT-STUD, BOLT NUT CRE AND NONFERROUS MATL (FOR FATIGUE OPNS, ISSUE CONTROLLED)
		AN126587 THRU AN126652	.375-16 X .374-24
		AN126881 THRU AN127192	NECKED, .375-16 X .375-24
		AN127193 THRU AN127492	NECKED, .4375-14 X .4375-20
		AN128363 THRU AN128686	DRILLED, .250-20 X .250-28
		AN128687 THRU AN128998	DRILLED, .3125-18 X .3125-24
		AN129293 THRU AN129604	DRILLED, NECKED, .375-16 X .375-24
		MS17299	CONTROLLED ROOT RAD, .190-24 X .190-32
		MS17300	CONTROLLED ROOT RAD, .250-20 X .250-28
		MS17301	CONTROLLED ROOT RAD, .3125-18 X .3125-24

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TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
STUDS, WELDING	152042	MIL-S-24149	STUDS, WELDING
		FF-T-791	TURNBUCKLE
		MS27951	TURNBUCKLE
STUD, SELF-CLINCHING	152043	MS21326	STUD, SELF-CLINCHING

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2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-S-85	Screw, Caps, Slotted and Hexagon Head
FF-S-86	Screw, Cap, Socket Head
FF-S-1362	Stud, Plain, General Purpose
QQ-P-416	Plating, Cadmium (Electrodeposited)

MILITARY

MIL-S-1222	Studs, Bolts, Hex Cap Screws, Socket Head Cap Screws and Nuts
MIL-B-7838	Bolt, Internal Wrenching, 160 KSI Ftu
MIL-B-7874	Bolt, Machine, 1200 Degree F
MIL-B-8831	Bolt, Tensile, Steel, 180 KSI Ftu, 450 Degree F, External Wrenching, Flanged Head
MIL-B-8906	Bolt, Tensile, Steel, 220 KSI Ftu, 450 Degree F External Wrenching, Flanged Head
MIL-B-8907	Bolt, Tensile, Steel, 260 KSI Ftu, 450 Degree F, External Wrenching, Flanged Head
MIL-B-45912	Bolt, Self-Locking, Heaxgon Head, Steel
MIL-F-8961	Fastener, Externally Threaded, 450 Degree F and 1200 Degree F Self-Locking Design
MIL-F-18240	Fastener, Externally Threaded, 250 Degree F Self-Locking Element For
MIL-I-17214	Indicator, Permeability, Low Mu (Go-No-Go)
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For

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MIL-S-45909	Stud, Locked In - Ring Lock Serrated
MIL-S-45919	Stud, Lock In - Key Locked
MIL-S-45933	Stud, Keyring Locked, 125 and 160 KSI, Ftu: General Specification For

STANDARDS

FEDERAL

FED-STD-151	Metal Test Methods

MILITARY

DOD-STD-35	Automated Engineering Document Preparation System
MIL-STD-130	Identification Marking of Military Property
MIL-STD-1312	Fasteners, Test Methods
MIL-STD-1598	Studs, Preferred For Design, Listing Of

(Copies of specifications, standards, drawings, and publications required by contractors, in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSIB46.1	Surface Texture (Surface Roughness, Waviness, and Lay)
ANSIB18.6.4	Thread Forming and Thread Cutting Tapping Screws and Metallic Drive Screws (Inch Series)

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

ASTMA370	Method and Definitions for Mechanical Testing of Steel Products
ASTME8	Tension Testing of Metallic Materials, Methods Of

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ASTME10	Method of Test for Brinell Hardness of Metallic Materials
ASTME18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Test For
ASTME92	Method of Test for Vickers Hardness of Metallic Materials
ASTME112	Estimating the Average Grain Size of Metals
ASTME380	Metric Practice Guide

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

NAS621	Fasteners, Titanium Alloy Procurement Specification
NAS4002	Fasteners, Alloy Steel, Externally Threaded
NAS4003	Fasteners, A286 CRES, Externally Threaded
NAS4004	Fasteners, 6AL-4V Titanium Alloy Externally Threaded

(Application for copies should be addressed to the Aerospace Industries Association of America, Inc., 1825 DeSales Street, N.W., Washington, D.C. 20036.)

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3. DEFINITIONS

Definitions contained in DOD-STD-35 are applicable to Bolts and Studs.

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4. GENERAL REQUIREMENTS

4.1 Introduction. In order to properly prepare an SRS for Bolts or Studs, it is necessary to use DOD-STD-35 in conjunction with this dash-numbered part. In the event the instructions or the writing methodology are not followed, an AEDPS edit routine will print error messages rather than the desired MSE document.

4.2 Specification Requirement Sheet (SRS). SRS requirements described in DOD-STD-35 are applicable to Bolts and Studs.

4.3 Configuration.

4.3.1 Dimensions. SRS requirements for dimensisions described in DOD-STD-35 are applicable to Bolts and Studs.

4.3.2 Illustrations. Illustrations applicable to Bolts and Studs are contained in Appendix A to this document.

4.4 Units of measure. The International System of Units (SI) or inch-pound units (U.S.), may be used to specify characteristics (e.g. weight, dimensions). Refer to ASTME380 for guidance in using SI. However, when different units of measure are available for specifying a characteristic, only one may be used. Both units of measure cannot appear in the same MSE.

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5. DETAILED REQUIREMENTS

5.1 Introduction. This section provides instructions for completion of the SRS and references for instructions on specific requirements contained in DOD-STD-35.

5.2 MSE formats. AEDPS has the capability to prepare a six-section or a single-section document. The six-section document appears similar to a six-section military specification. The single-section document appears similar to a military specification sheet. Specification requirements described in DOD-STD-35 are applicable to either of the formats selected. It is mandatory that one of the two following codes and an "X" be entered on the SRS to create the desired MSE format:

a. 0700 for a six-section format

b. 0800 for a single-section format

5.3 Referenced documents. Section 2 of the MSE may contain a list of all referenced documents within Sections 3, 4, and 5 of the MSE. Instructions for referencing documents to appear in Section 2 of the forthcoming MSE are contained in DOD-STD-35.

5.4 Deleted

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5.5 Tabulation. Instructions for tabulation are provided in DOD-STD-35. Table III provides codes for column headings which may be used for bolts and studs characteristics. When different units of measure within one system (U.S. or SI) are available for specifying a characteristic, only one of the units may be tabulated; e.g. either ounces or pounds.

TABLE III. TABULATION CANDIDATES

CODE	COLUMN HEADING	CODE	COLUMN HEADING	CODE	COLUMN HEADING
1771	THD DES	1773	THD DES END 1	1775	THD DES END 2
1777	INTL THD DES	2795	ULT TNSL STR KSI	2797	ULT TNSL STR MPA
2799	TNSL YIELD STR KSI	2801	TNSL YIELD STR MPA	7044	WT PER 100 KILOGRAMS
2805	PROOF STR POUNDS	2807	PROOF STR NEWTONS	7042	WT PER 100 POUNDS
7041	WT PER 100 OUNCES	7021	WT MAX EA GRAMS	7022	WT MAX EA OUNCES
7023	WT MAX EA POUNDS	6996	WT NOM EA OUNCES	6997	WT NOM EA POUNDS
6998	WT NOM EA GRAMS	6999	WT NOM EA KILOGRAMS	1778	SIZE DES
1769	THREAD DOC DES				

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CHAR CODE	STANDARD PARAGRAPH
6996	NOMINAL WEIGHT PER UNIT. THE WEIGHT PER UNIT SHALL BE 6996 OUNCES(S).
6997	NOMINAM WEIGHT PER UNIT. THE WEIGHT PER UNIT SHALL BE 6997 POUND(S).
6998	NOMINAL MASS PER UNIT. THE MASS PER UNIT SHALL BE 6998 GRAM(S).
6999	NOMINAL MASS PER UNIT. THE MASS PER UNIT SHALL BE 6999 KILOGRAM(S).
2297	BENDING. THE REQUIREMENT(S) SHALL BE IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 2297.
2299	BENDING. THE UNTHREADED SHANK SHALL NOT CRACK OR FRACTURE.
2281	BENDING. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 2281.
2283	BENDING. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH MIL-B-857.

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KEYWORD CODE INDEX (KCI)

CHARACTERISTIC NAME	CHAR CODE
CARBURIZATION AND DECARBURIZATION	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	2645
AS FOLLOWS:	2633
MAXIMUM DEPTH (SELECT ONE)	
(INCHES)	2651
(MILLIMETERS)	2653
MAXIMUM PERCENT OF SECTIONAL THICKNESS	
(PERCENT)	2655
MAXIMUM LIMITS, PER MIL-B-8907.	2657
MAXIMUM LIMITS, PER MIL-S-1222.	2658
TEST METHODS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	2662
PER DOCUMENT (DOC DES)	2664

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CHAR CODE	STANDARD PARAGRAPH
2645	CARBURIZATION AND DECARBURIZATION. THE REQUIREMENT(S) SHALL BE IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 2645.
2633	CARBURIZATION AND DECARBURIZATION. THE REQUIREMENT(S) SHALL BE AS FOLLOWS:
2651	THE DECARBURIZED DEPTH SHALL BE NOT GREATER THAN 2651 INCH(S).
2653	THE DECARBURIZED DEPTH SHALL BE NOT GREATER THAN 2653 MILLIMETER(S).
2655	THE PERCENT OF THE SECTIONAL THICKNESS SHALL BE NOT GREATER THAN 2655 PERCENT.
2657	THE DECARBURIZATION OF THE THREAD AREA SHALL BE NOT GREATER THAN THE LIMITS SHOWN IN MIL-B-8907.
2658	THE DECARBURIZATION OF THE THREAD AREA SHALL BE NOT GREATER THAN THE LIMITS SHOWN IN MIL-S-1222.
2662	CARBURIZATION AND DECARBURIZATION. TESTING SHALL BE PERFORMED IN ACCORDANCE WITH THE BASE DOCUMENT, EXCEPT 2662.
2664	CARBURIZATION AND DECARBURIZATION. TESTING SHALL BE IN ACCORDANCE WITH 2664.

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KEYWORD CODE INDEX (KCI)

CHARACTERISTIC NAME	CHAR CODE
CHEMICAL ANALYSIS	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	2666
AS FOLLOWS:	2668
SUPPLIER CERTIFICATION.	2672
TEST METHODS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	2674
PER DOCUMENT (DOC DES)	2676
METHOD (NUMBER)	2678
DUCTILITY	
REQUIREMENTS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	1939
MINIMUM ELONGATION (PERCENT)	1941
TEST METHODS (SELECT ONE)	
PER BASE DOCUMENT EXCEPT ____	1921

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